

Avionics Troubleshooting:

Magnetic Interference



Introduction & Objectives:

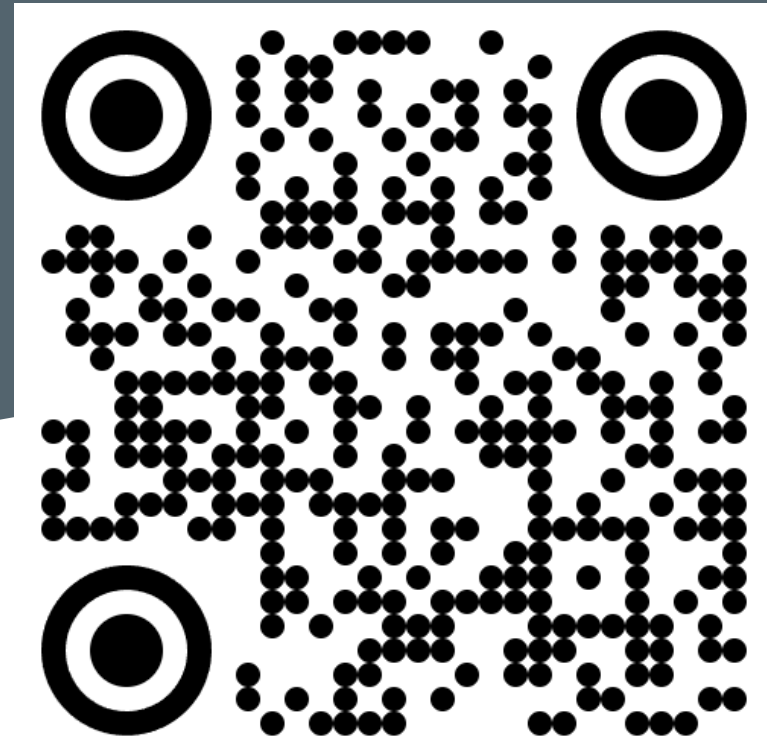
1. Goal: Reinforce key concepts from the 2024 troubleshooting presentation.
2. Focus on strategy, 7-step method, and practical application.
3. Understanding of magnetic interference and its impact on aircraft systems.
4. Troubleshooting Scenarios

STRATEGY OF TROUBLESHOOTING

Methods, techniques, & thought processes for troubleshooting technical issues

* Based on *The Art of Troubleshooting* by Jason Maxham © 2024

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Strategy of Troubleshooting

Strategy of Troubleshooting

- Troubleshooting is a learnable skill.
- Understand how people describe problems.
- Use psychology and empathy to get into the operator's mindset.

Strategy of Troubleshooting

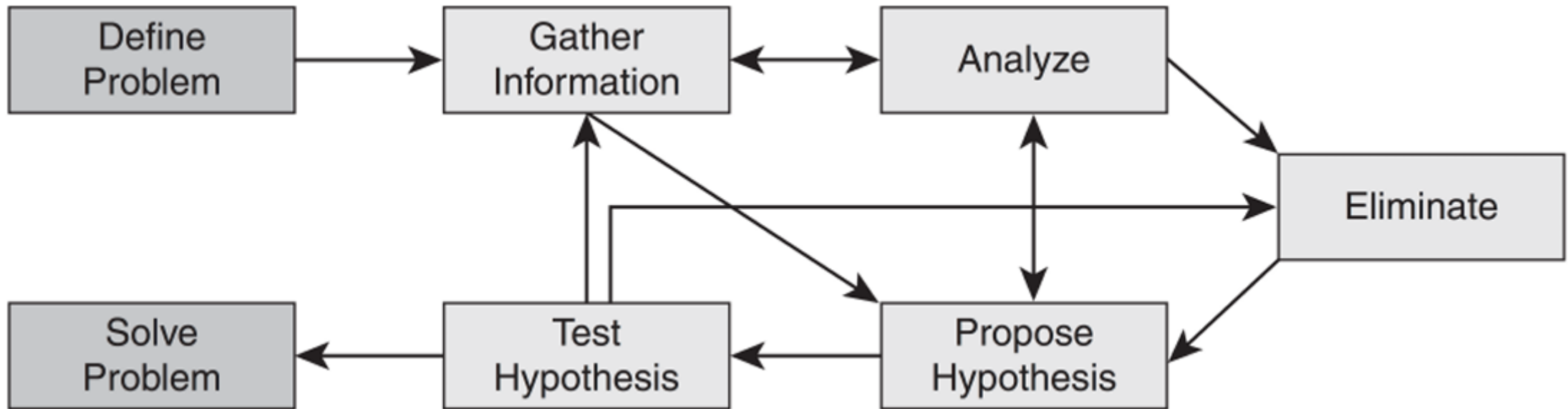
- Consider economic and organizational context.
- Know your own strengths and limitations.

7-Step Method of Troubleshooting

7-Step Method of Troubleshooting

1. **Define the Problem**
2. **Gather Facts**
3. **Analyze Information**
4. **Eliminate Possibilities**
5. **Propose a Hypothesis**
6. **Test the Hypothesis**
7. **Solve the Problem**

7-Step Method of Troubleshooting



Challenge

1. **What's the first step in troubleshooting?**

Challenge

1. **What's the first step in troubleshooting?**
2. **Why change only one thing at a time when testing?**

Challenge

- 1. What's the first step in troubleshooting?**
- 2. Why change only one thing at a time when testing?**
- 3. What should always be done after a repair?**

Strategy of Troubleshooting

- Troubleshooting is a process, not a guess.
- Clear communication and methodical thinking are key.
- Always retest and consider system-wide impacts.

Magnetic Interference

What is magnetic interference?

Magnetic interference in aircraft primarily affects the magnetometer/compass readings and can also impact other sensitive electronics.



What is magnetic interference?

The main source of magnetic interference is aircraft components, especially those with electrical currents or ferromagnetic materials, as well as lightning strikes.



What is magnetic interference?

This interference can lead to misleading heading data, heading failures, deviation, turning errors, and even circling in some flight modes.



Sources Of Magnetic Interference

Sources of Magnetic Interference

Permanent magnetization

Hard iron materials used in aircraft construction, like steel alloys in landing gear or engine mounts, can become permanently magnetized during manufacturing due to vibrations and the earth's magnetic fields



Sources of Magnetic Interference

Induced magnetization

Soft iron materials, such as aluminum alloys in the fuselage and wings, can temporarily become magnetized when exposed to external magnetic fields from electrical systems, other aircraft components, or the earth's magnetic field itself.



Sources of Magnetic Interference

Electrical Currents

Alternators, generators, electrical wiring, and on-board electronic systems generate magnetic fields, which can interfere with sensitive equipment.



Sources of Magnetic Interference

Lightning Strikes

While less common, lightning strikes can induce magnetism in steel hardware.



Sources of Magnetic Interference

External Magnetic Anomalies

Natural deviations in the earth's magnetic field (magnetic variation and local anomalies) and human-made sources like steel structures, power lines, and even electronic devices can distort the magnetic field.



Sources of Magnetic Interference

Aircraft components

Electronics - radios, navigation systems and other electrical devices can generate electrical fields that interfere with heading readings.



Sources of Magnetic Interference

Hardware /Airframe

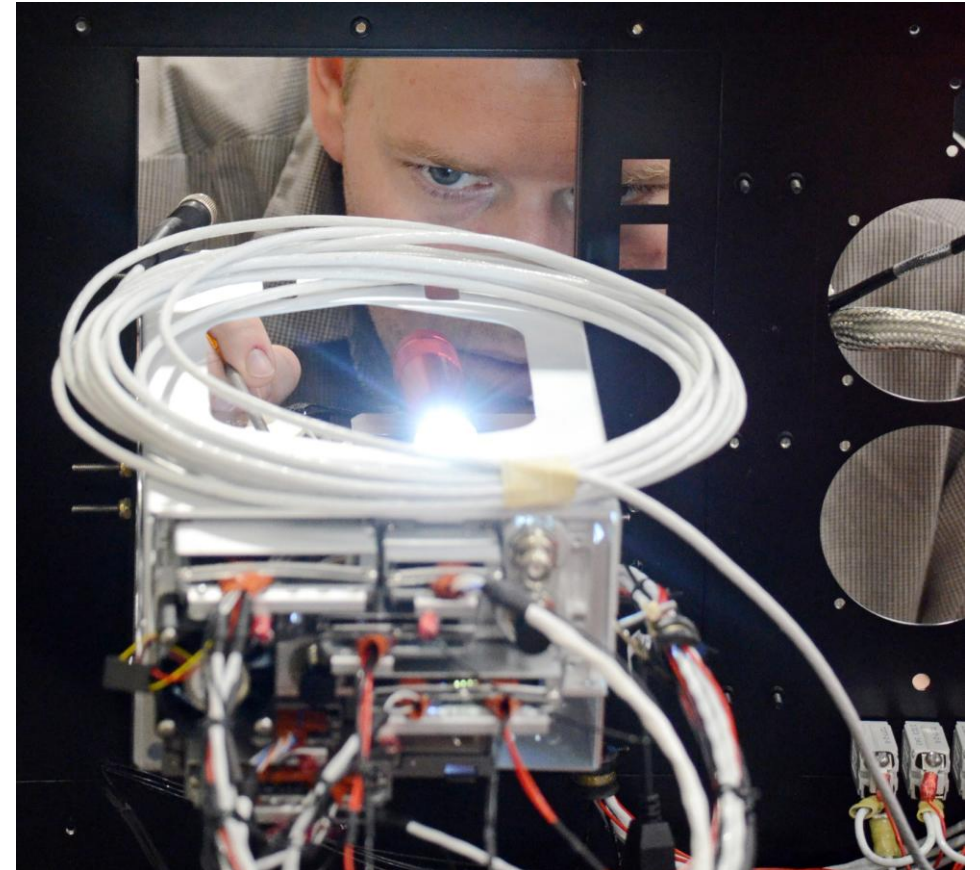
Ferro magnetic materials, like steel, can become magnetized and cause deviations in the magnetic heading in the magnetometer/compass.



Sources of Magnetic Interference

Wires and Cables

Electrical wires, especially those carrying high currents near equipment, can induce magnetic fields that affect the magnetometer/compass.

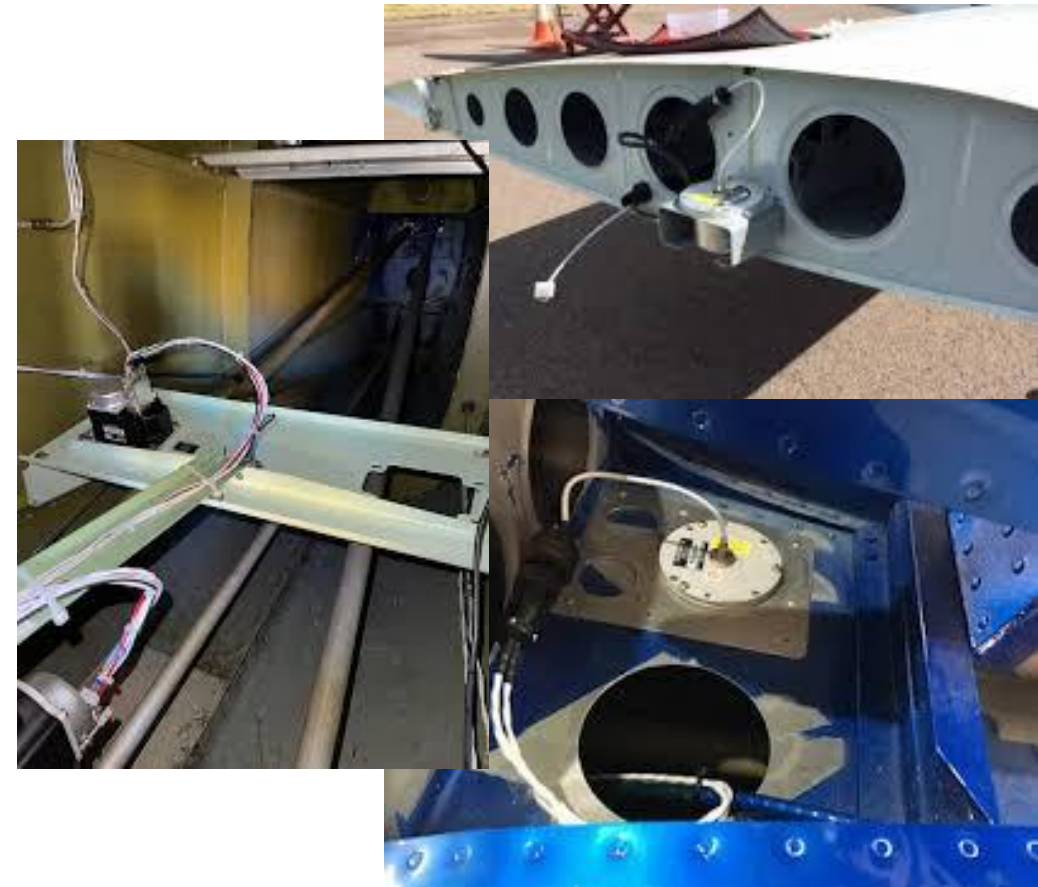


Mitigation & Correction of Magnetic Interference

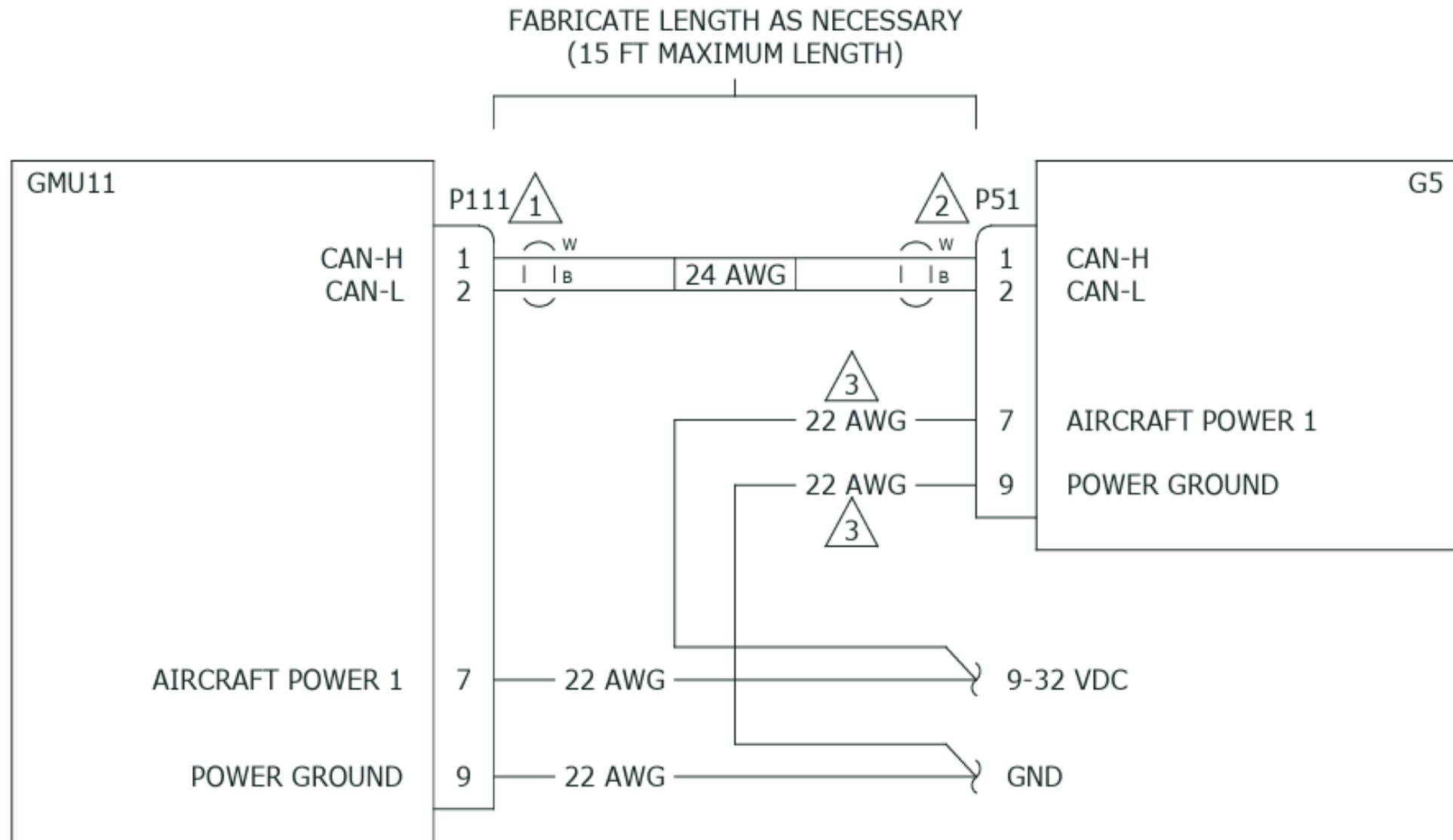
Mitigation & Correction of Magnetic Interference

Magnetic interference survey

Always perform a magnetic interference survey of the proposed installation site of your magnetometer before installing components.



Shop fabricated G5 magnetic survey tool



Example Magnetic Survey Test Sequence

A/C Reg.:_____		Survey data file name:_____
Elapsed Time (sec)	Elapsed Time (min:sec)	Action
0	0:00	Test begins (calibration period – no activity permitted)
20	0:20	Calibration period ends
30	0:30	Navigation lights on
40	0:40	Navigation lights off
50	0:50	Landing lights on
60	1:00	Landing lights off
70	1:10	Taxi lights on
80	1:20	Taxi lights off
90	1:30	Air conditioning on
100	1:40	Air conditioning off
110	1:50	Landing lights and taxi lights on
120	2:00	Landing lights and taxi lights off
140	2:20	Strobes on
160	2:40	Strobes off
180	3:00	Recognition lights on
190	3:10	Recognition lights off
200	3:20	Beacon on
210	3:30	Beacon off
220	3:40	AP engaged in a pitch and roll made (to engage servos)
230	3:50	AP disengaged
240	4:00	Pitot heat on
250	4:10	Pitot heat off
260	4:20	End of test

Mitigation & Correction of Magnetic Interference

Incandescent Nav lights

Major cause of interference and is very common because the lights ground return is connected to the airframe at the lamp location.



Mitigation & Correction of Magnetic Interference

Incandescent Nav lights – Good Practice

- Run shielded power and grounds to the light's location and terminate the ground at the fuselage so there is no ground return current in the wing.
- Sometimes you may have to float/insulate the unit, so it doesn't have a ground return path through the wing.

Mitigation & Correction of Magnetic Interference

Incandescent Nav lights

Replacing with new low current LED versions are really good to reduce the current draw and usually fix the problem.

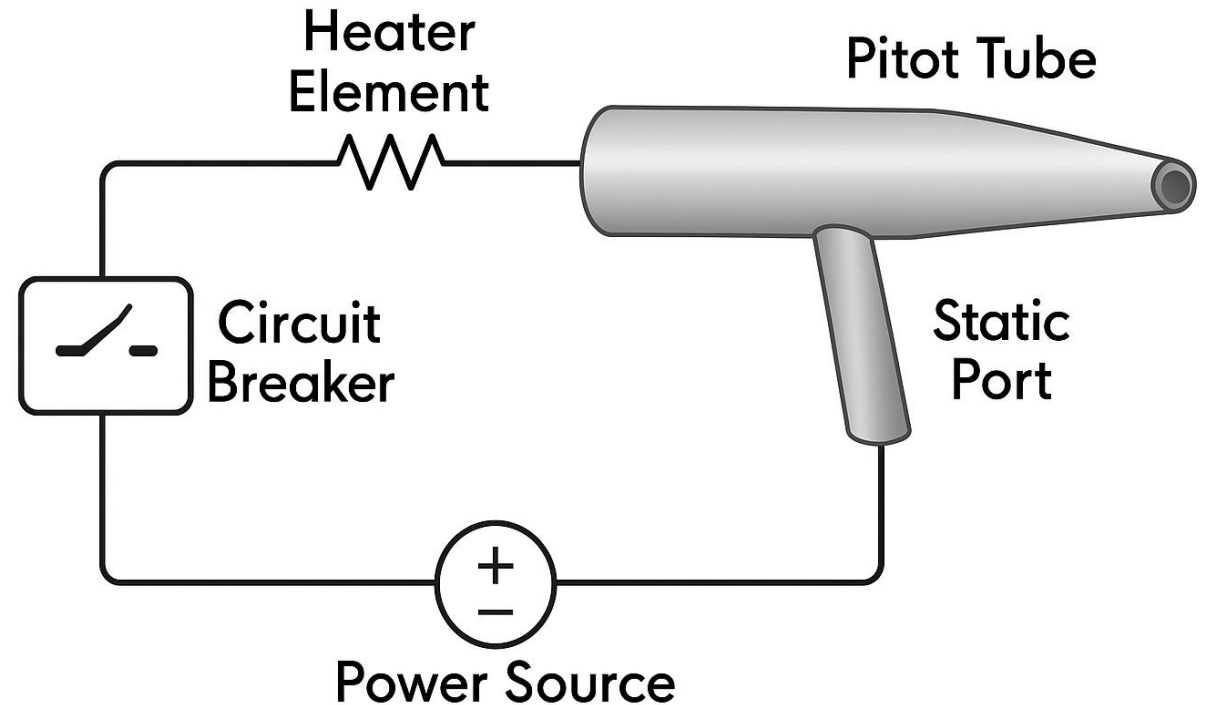


Mitigation & Correction of Magnetic Interference

Pitot heat systems

Improper power or grounding connections for pitot heat systems may result in electrical interference.

Pitot Heat System



Mitigation & Correction of Magnetic Interference

Pitot heat systems – Good Practice

- Install the magnetometer as far outboard as possible from the pitot tube.

Recommended distance from sources of interference

Source of Magnetic Interference in Order of Importance	Min. Distance Feet [meters]
1. Electric relays and motors, including servo motors	10.0 [3.0]
2. Ferromagnetic (i.e., iron, steel, or cobalt materials) mass heavier than 1 kg (e.g. landing gear)	8.2 [2.5]
3. Ferromagnetic mass less than 1 kg total (e.g., control cables)	3.0 [1.0]
4. Electrical devices that draw a current of more than 100 mA	3.0 [1.0]
5. Electrical conductors that pass a current of more than 100 mA	3.0 [1.0]
6. Electrical devices that draw a current of less than 100 mA	2.0 [0.6]
7. Magnetic measuring device other than another GMU (e.g., existing flux gate, even if not powered)	2.0 [0.6]
8. Electrical conductors passing less than 100 mA current	1.3 [0.4]

Mitigation & Correction of Magnetic Interference

Things that can help

- Hardware change.
- Degaussing structures/airframe.
- Move magnetometer to a clean location.
- Replacing existing wiring, shielding, move ground returns.
- Update position lights to LED model.

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